Cyclistic
Bike-Share

How Casual Riders and Annual Members Use Cyclistic Differently

A data-driven strategy to understand usage patterns and convert casual riders into annual members.

- **PREPARED BY**
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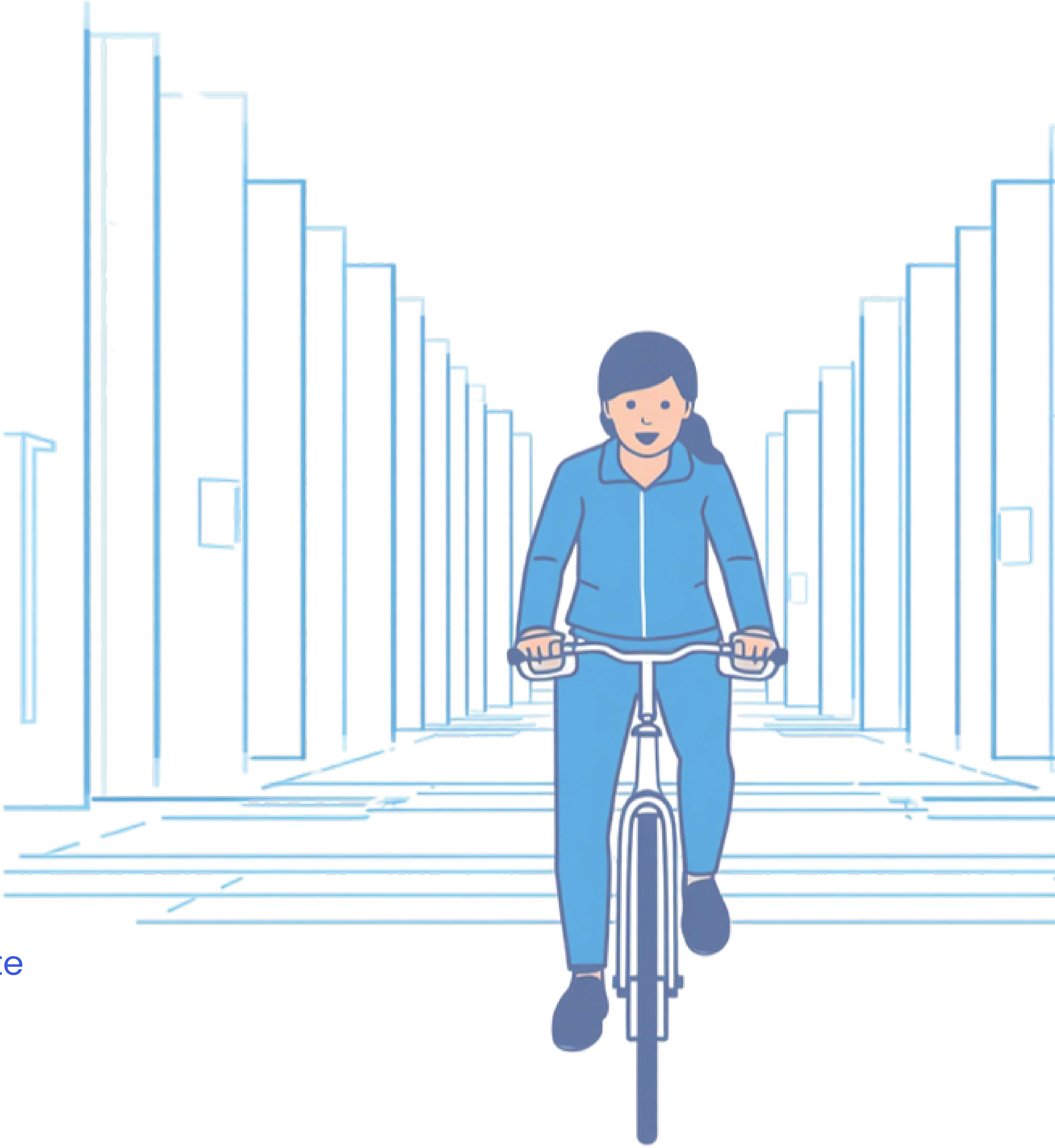
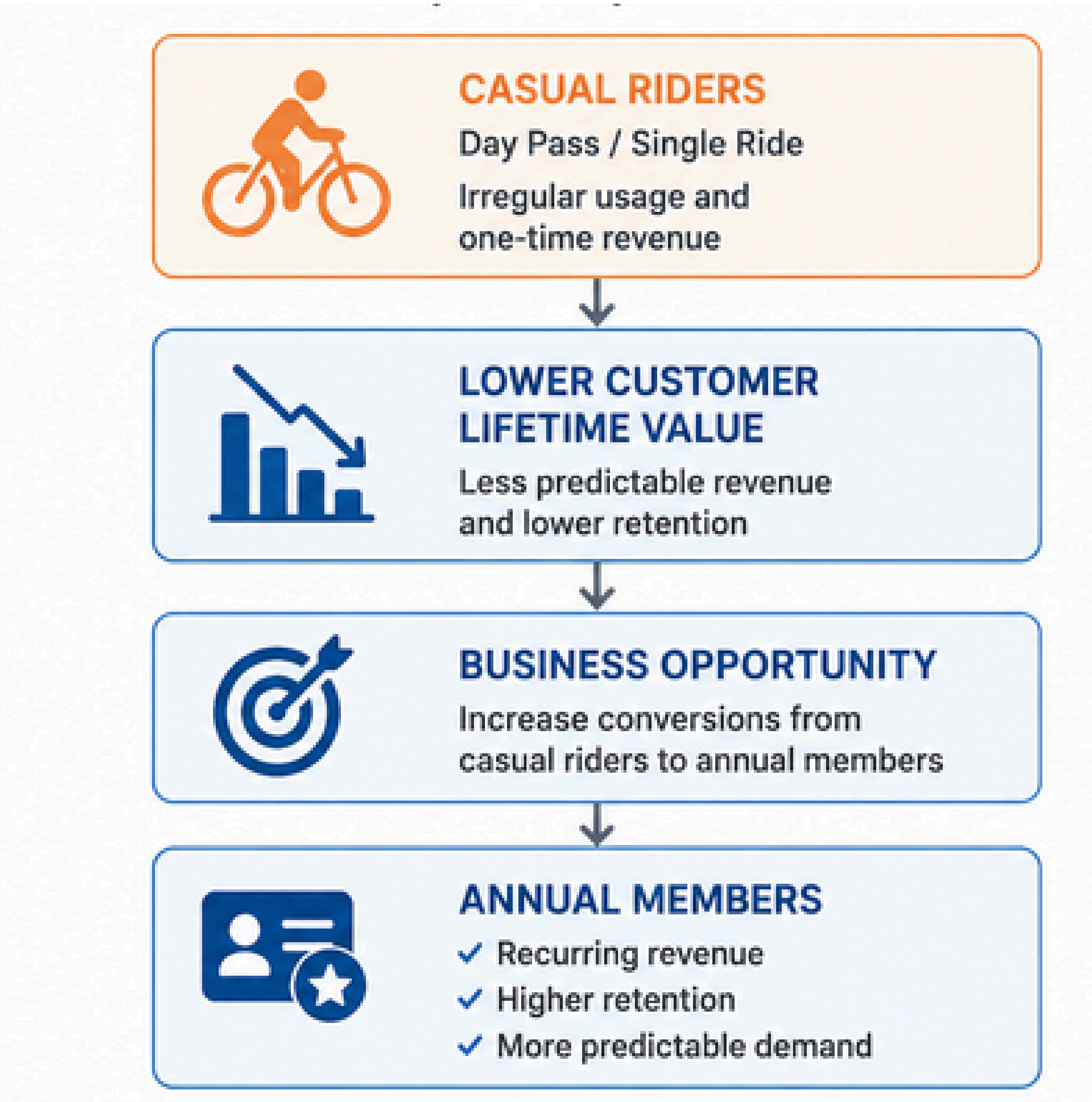
- **PROJECT**
  Google Data Analytics Professional Certificate



2026
Analysis

Today's Agenda

Presentation structure and key discussion points for the Cyclistic Data Analytics Capstone ↗



Business Problem

CYCLISTIC NEEDS TO INCREASE ANNUAL MEMBERSHIPS FOR SUSTAINABLE GROWTH ↗

Annual members generate more long-term value than casual riders, making membership conversion a strategic business priority.

Business Challenge

Cyclistic understands that annual members are more valuable, but lacks evidence-based insights into how casual riders differ from members and what motivates conversion.

Business Task

3

HOW DO CASUAL RIDERS AND ANNUAL MEMBERS USE CYCLISTIC DIFFERENTLY?

The analysis aims to identify meaningful behavioral differences that can support marketing strategies designed to increase annual memberships.

Key Questions

The analysis focuses on identifying differences in:

- Riding frequency
- Ride duration
- Ride distance
- Time of day
- Day of week
- Seasonal patterns
- Bike preferences
- Popular starting stations

Expected Outcome

Deliver actionable insights that help Cyclistic design targeted marketing campaigns to convert casual riders into annual members.

Cyclistic Historical Trip Data

JUNE 2025 – MAY 2026

KEY VARIABLES

- Ride ID
- Rideable Type
- Start & End Time
- Start & End Station
- User Type

TOOLS USED

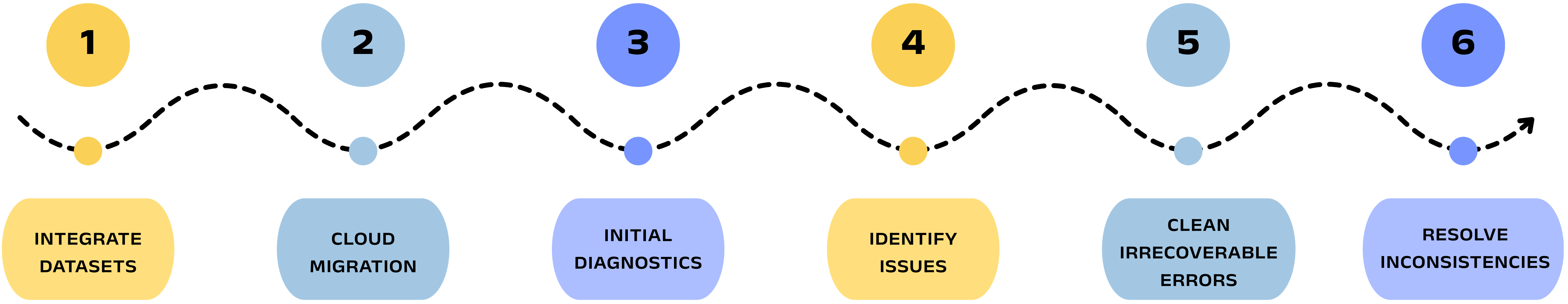
- SQL
- Python (Pandas)
- Tableau Public

5.68 M

CASUAL RIDERS & ANNUAL MEMBERS

Preparing High-Quality Data for Reliable Analysis

CLEANING AND TRANSFORMING THE DATA ENSURES THAT SUBSEQUENT ANALYSIS IS ACCURATE, CONSISTENT, AND TRUSTWORTHY.



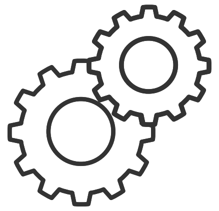
PREPARE & DIAGNOSE

- ✓ Downloaded 12 months trip data
- ✓ Checked column types, formats, missing values
- ✓ Verified time ranges & data integrity (5.85M rows)
- ✓ Identified duplicates, abnormal durations, nulls, inconsistent station IDs/names

CLEAN DATA

- ✓ Dropped irrecoverable errors (duplicates, negative/short/long trips)
- ✓ Resolved station ID/name mismatches via GPS coordinates
- ✓ Standardized station identifiers
- ✓ Produced final clean dataset (~5.68M trips) ready for analysis

01

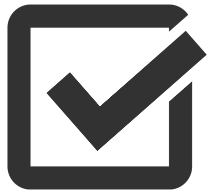


FEATURE ENGINEERING

Create analysis-ready variables that describe rider behavior:

- **ride_length** – trip duration (seconds → minutes for reporting)
- **day_of_week** – readable weekday labels
- **distance_km** – straight-line distance calculated using the Haversine formula

02



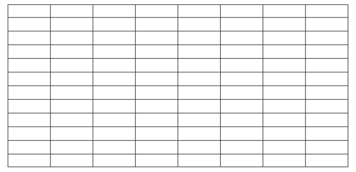
DATA VALIDATION

Validation includes checking:

- data types
- missing values
- unrealistic durations
- distance calculations
- weekday formatting

Reliable analysis depends on verified data rather than assumptions.

03



SUMMARY TABLES

Summary tables were created for:

- Overall summary
- Ride duration
- Ride distance
- Ride count by weekday
- Ride count by hour
- Ride count by month
- Bike type preference
- Top start stations

04

BEHAVIORAL PATTERNS

Key comparison dimensions include:

- Riding frequency
- Ride duration
- Ride distance
- Time of day
- Day of week
- Seasonality
- Bike preference
- Popular start stations

Why Summary Tables?

Aggregating the dataset before visualization reduces computational overhead, improves reproducibility, and ensures that every chart in the Tableau dashboard is directly traceable to a validated analytical query.

Overall Comparison

Total Rides

3.67 M Member | **2.01 M** Casual

Member completed **1.8×** more rides

Loop Trips

2.6 % Member | **7.2 %** Casual

Casual is nearly **3× higher**

Average Ride Duration

Member **12.3 min** < Casual **19.6 min**

59%

Average Ride Distance

Member **2.28 km** ~ Casual **2.23 km**

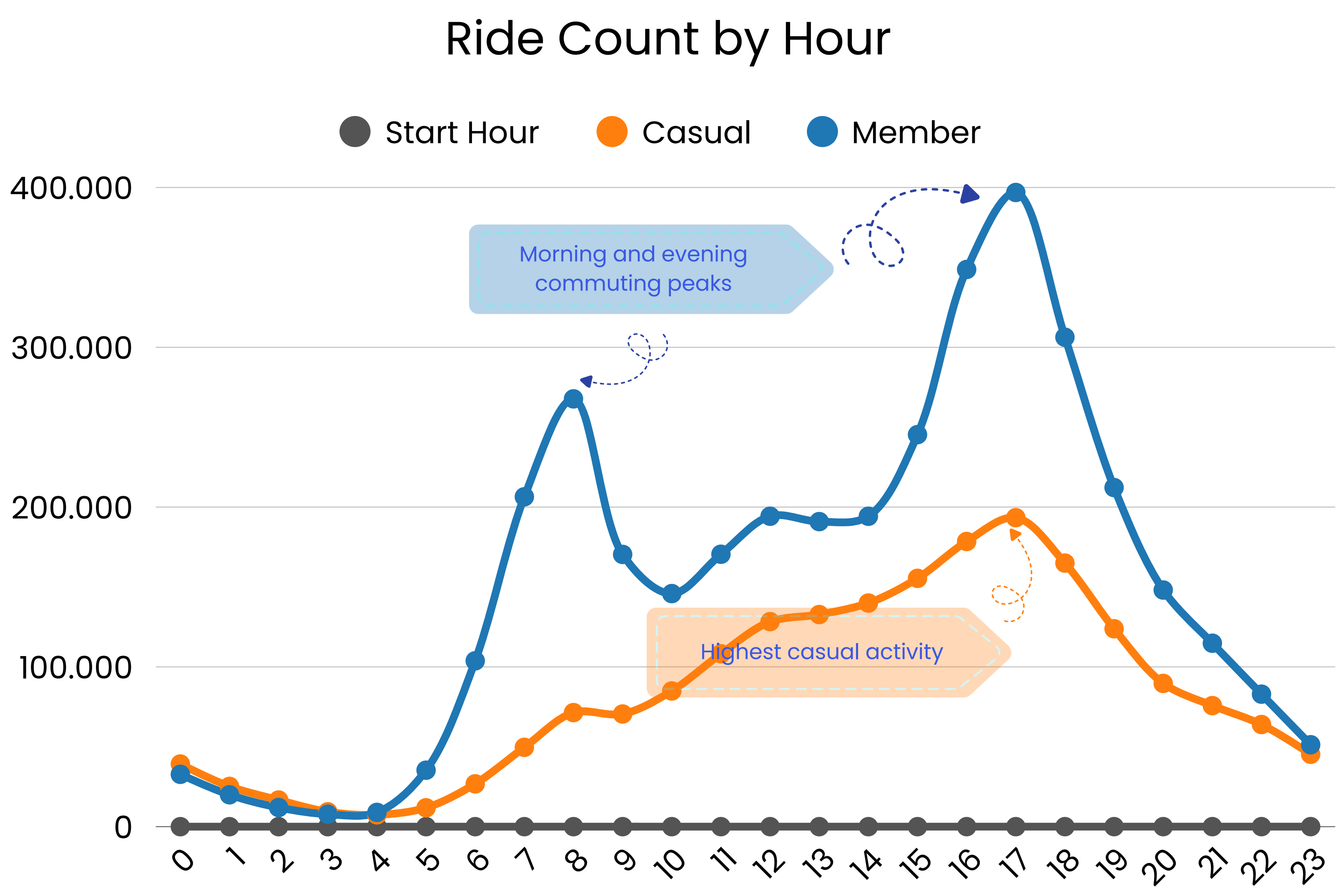
Almost identical

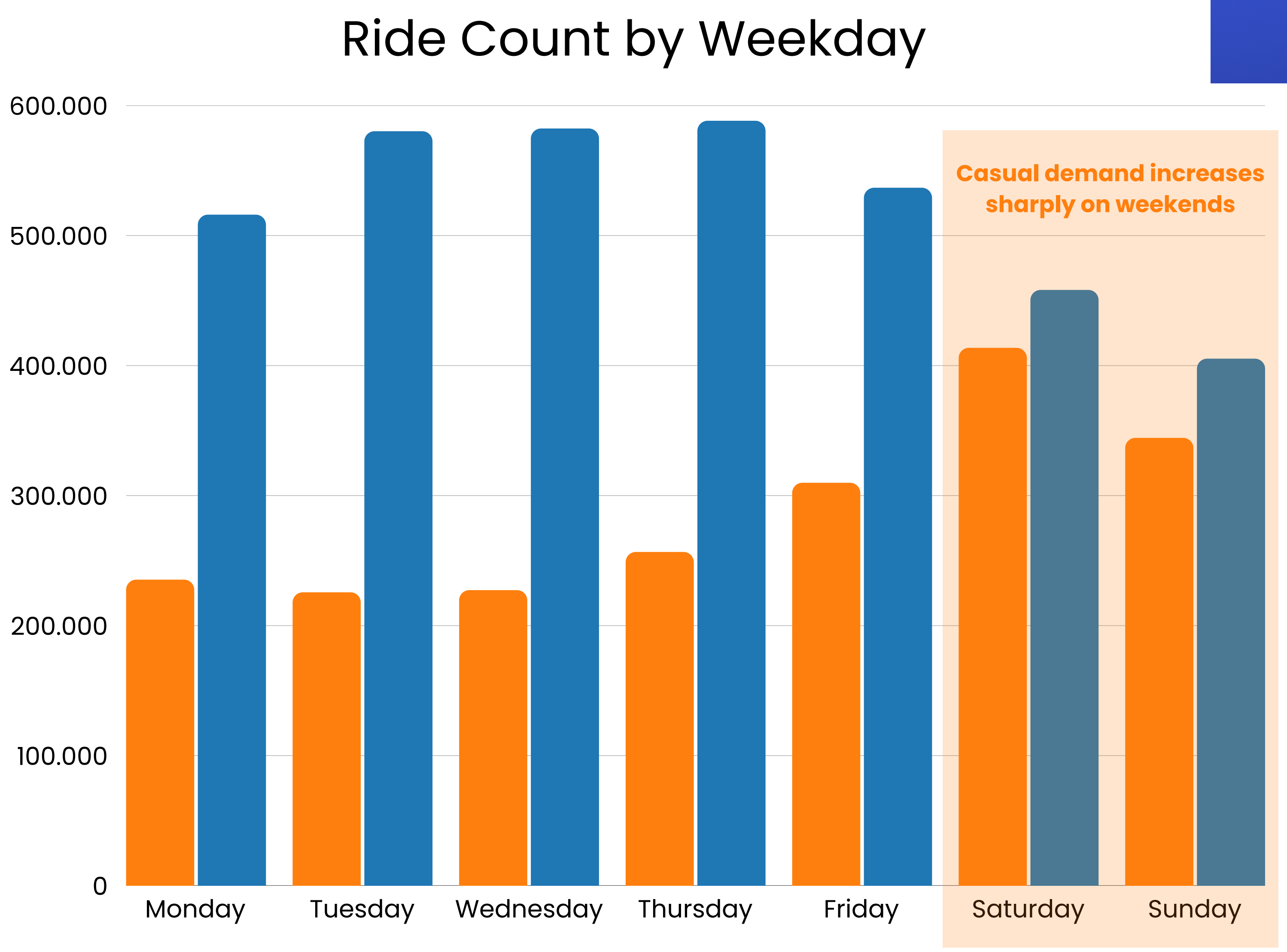
KEY TAKEAWAYS

- Casual riders **take fewer trips** but **spend significantly longer on each ride**, while traveling nearly **the same distance** as annual members.

WHY THIS MATTERS

- These differences suggest that **casual riders primarily use Cyclistic for leisure**, whereas **annual members rely on it for everyday transportation**.





Daily Usage Patterns

KEY TAKEAWAYS

- Annual members exhibit two distinct commuting peaks during weekday rush hours.
- Casual riders show no pronounced morning peak and ride more frequently during afternoons.
- Member usage remains stable across weekdays, while casual ridership increases substantially on weekends.

WHY THIS MATTERS

- Hourly and weekly patterns clearly indicate that annual members rely on Cyclistic for commuting, whereas casual riders primarily use it for recreational trips.

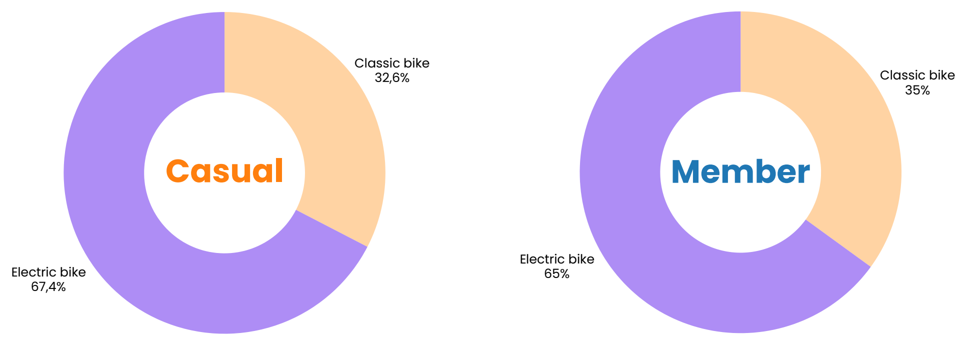
Supporting Behavioral Evidence

KEY TAKEAWAYS

- Both rider groups follow similar seasonal trends, but casual riders concentrate around tourist destinations while annual members begin trips in business districts.

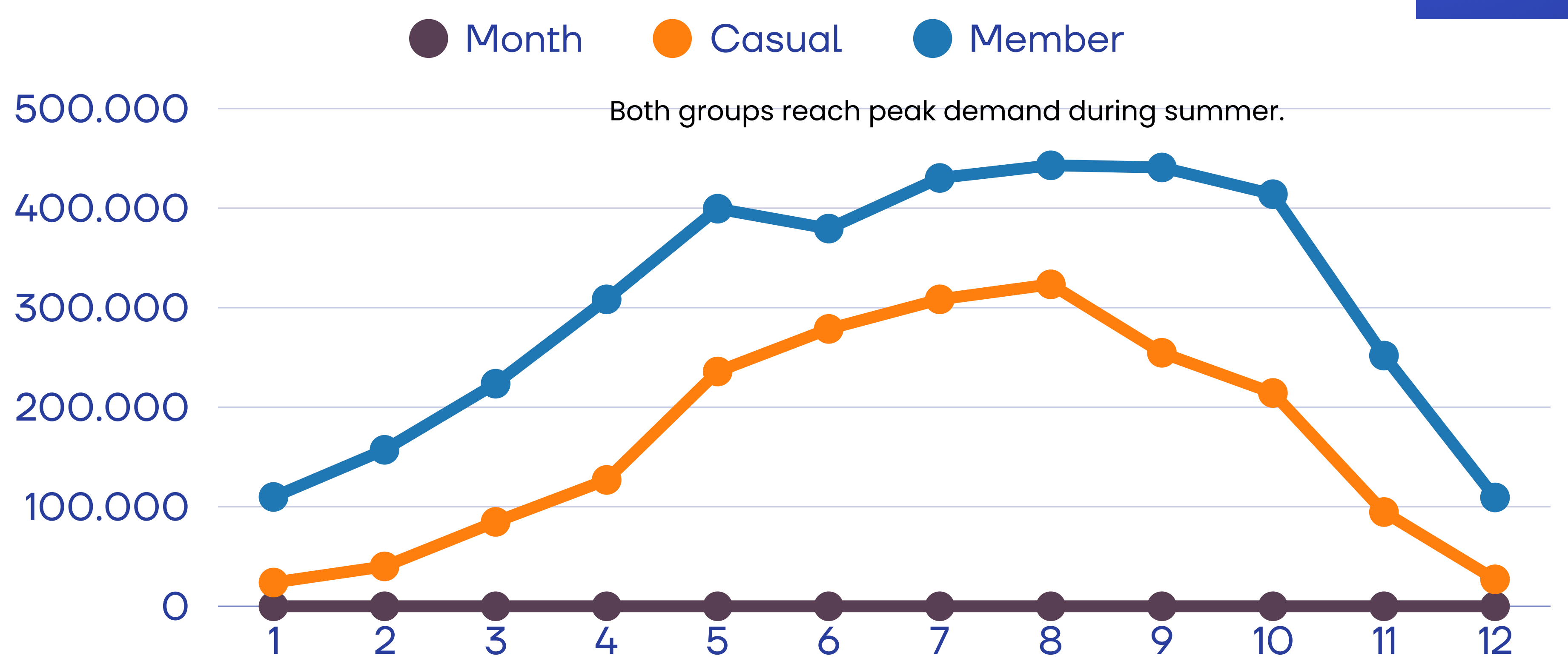
WHY THIS MATTERS

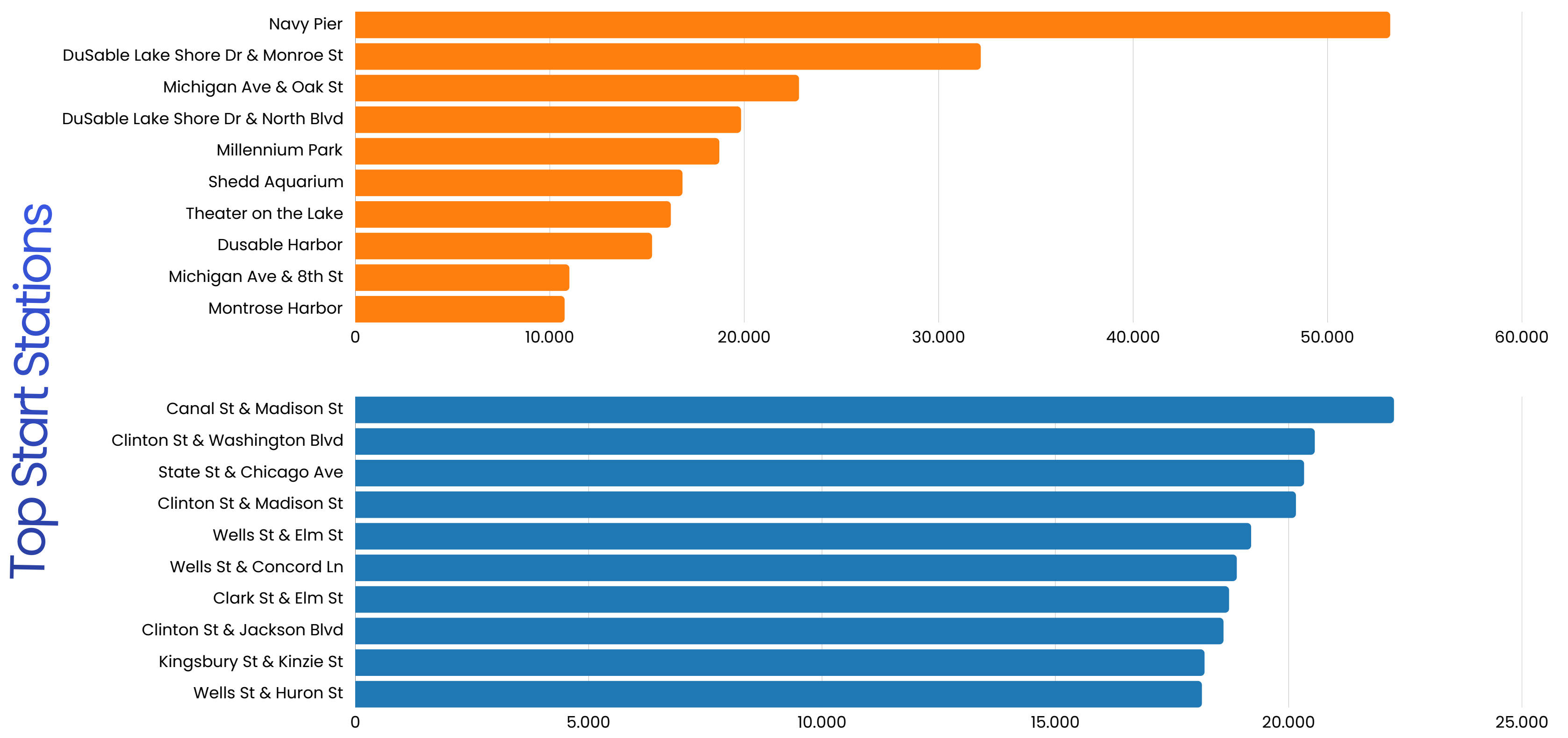
- Seasonality indicates when demand is highest, while starting locations provide additional evidence of different travel purposes.



Bike Preference

Ride Count by Month





Business Recommendations

BEHAVIORAL INSIGHTS SUGGEST THAT MEMBERSHIP CONVERSION SHOULD FOCUS ON THE RIGHT CUSTOMERS, AT THE RIGHT TIME, AND IN THE RIGHT LOCATIONS.

01

SEASONAL CAMPAIGNS

13× from January to August

Launch membership campaigns between April and June, using limited-time discounts or free trial memberships before peak riding season.

>> Reach casual riders when engagement is highest and increase membership conversion.

02

POSITION MEMBERSHIP FOR LEISURE RIDERS

59% longer per trip

Position annual membership as the best value for frequent leisure riders, emphasizing unlimited rides, convenience, and long-term savings.

>> Increase perceived value among riders who already use Cyclistic for extended recreational trips.

03

TARGET HIGH-TRAFFIC LEISURE LOCATIONS

Navy Pier, Millennium Park, and DuSable Lake Shore Drive

Deploy QR-code sign-ups, digital advertising, and seasonal membership promotions at major recreational destinations.

>> Reach potential members at the locations where they naturally begin their rides.

04

STRENGTHEN THE COMMUTER VALUE PROPOSITION

Consistent weekday commuting patterns

Promote annual membership as a reliable commuting solution by highlighting unlimited access, convenience, and cost efficiency.

>> Encourage frequent riders to transition from occasional use to annual membership.

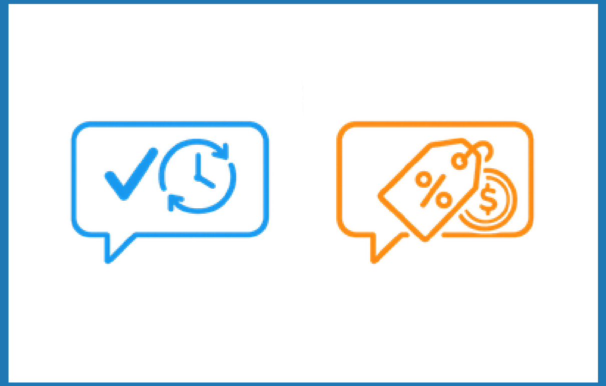
Strategic Recommendations

TURNING BEHAVIORAL INSIGHT INTO A TARGETED CONVERSION STRATEGY

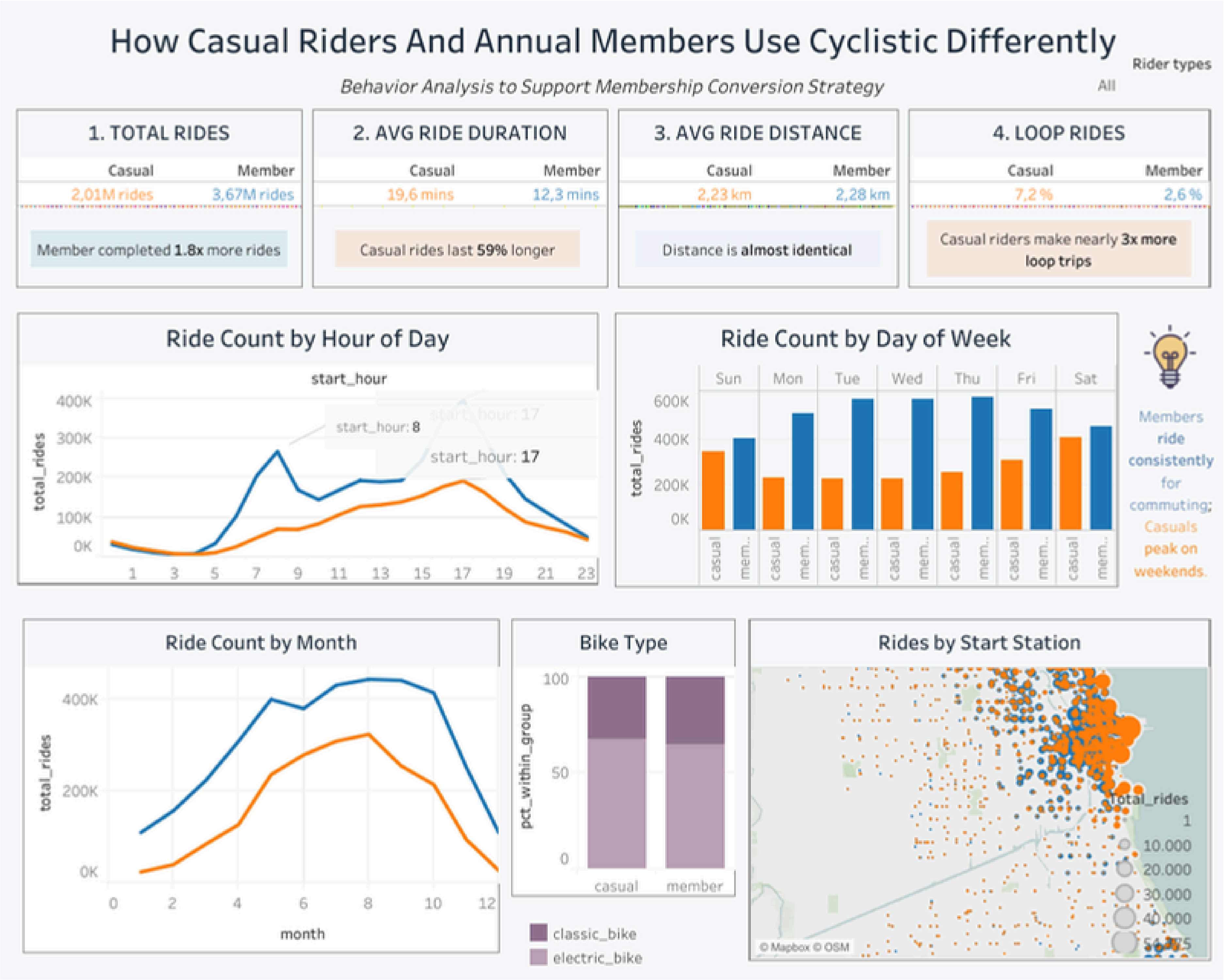
Rather than targeting all casual riders equally, Cyclistic should prioritize high-engagement leisure riders — during peak season, at high-traffic recreational locations — where conversion potential is highest.

Different Purpose

Different Timing



Different Approach



Dashboard Tableau

Interactive Dashboard

EXPLORE THE INTERACTIVE DASHBOARD FOR DEEPER ANALYSIS. ↗

Next Steps

VALIDATING RECOMMENDATIONS AND EXPANDING ANALYSIS

Data-driven decision making is an iterative process. Behavior should be monitored continuously to refine marketing strategies over time.

Monitor Business Impact

Evaluate whether the proposed marketing strategies successfully increase annual membership conversion.

- Membership conversion rate: Percentage of casual riders who become annual members
- Campaign performance: Compare conversion rates across seasonal promotions, free trials, and limited-time discounts.
- Customer retention: Measure how many new members renew after their first year.
- Seasonal demand: Compare ridership patterns before and after campaign implementation.

Expand Future Analysis

Additional data that would strengthen the analysis

- Rider demographics: Age, Gender, Resident vs Tourist, Income (if available)
- Membership history: How many rides occur before upgrading?, Average time to convert?
- Weather: Temperature, Rainfall, Wind, Could explain seasonal changes.
- Marketing exposure: Which users actually saw, QR campaigns, Email campaigns, Promotions, Without this, we cannot measure campaign effectiveness.

Thank You!
Questions?
Let's connect and discuss!

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Cyclistic
Analytics

This analysis was conducted as part of the Google Data Analytics Professional Certificate Capstone Project. All insights are data-driven and designed to support Cyclistic's marketing strategy for converting casual riders into annual members.

2026
Capstone